

PRISM-ANT Algorithm Specification

Version 1.0 | Predictive, risk-informed, safety-masked station allocation

Purpose. PRISM-ANT is the station allocation kernel. It converts model outputs and live EV constraints into safe power caps, an auditable priority, and a full-utilization allocation. The implementation is designed to explain every kW allocated.

1. Operating sequence

1. Acquire. Receive EV telemetry, requested target/deadline, emergency status, charger and site capacity. **2. Infer.** Run M1-M5 and retrieve the M6 candidate-policy value. **3. Safety-mask.** Compute the feasible cap before any priority calculation. **4. Score.** Compute a session priority only for safe candidates. **5. Allocate.** Use capped water-filling until site capacity is exhausted or every safe cap is reached. **6. Re-run.** Repeat on arrival, completion, temperature change, or emergency update.

2. Hard safety mask

The feasible cap is the minimum of vehicle intake, charger/site electrical limits, current-temperature cap, M4 recommended power envelope, M5 future-temperature trajectory cap, and - for normal sessions - an M3 health-protection cap. A score cannot override this result.

Emergency mode: emergency status removes only the normal battery-life preference. It does not remove current-temperature, future-temperature, vehicle, charger, or site limits.

3. PRISM priority formula

For a feasible session i , the implementation uses: $P_i = .30 U + .24 R + .18 N + .10 Q + .08(1-F) + .05(D \times R) - .06 H + E$. U is urgency, R is departure-readiness risk, N is normalized energy need, Q is M6 strategy value, F is flexibility, D is M2 forecast pressure, H is M3 health protection, and E is the emergency increment. The $D \times R$ interlock prevents forecast load from acting as an identical, non-discriminating bonus for every session.

4. Model-to-PRISM signal contract

Model	Input class	Output passed to PRISM	How it is consumed
M1	SOC, capacity, target, route, deadline	energy_need_kwh; confidence; readiness_risk	N and R raise service priority; confidence is logged.
M2	Load, queue, capacity, horizon	forecast_pressure	D x R raises urgency only for deadline-risk sessions during forecast pressure.
M3	SOH, cycles, temperature, resistance	health_risk; health_confidence	H protects normal-session battery life and may lower a normal cap.
M4	Temperature, C-rate, requested power	thermal_risk; safe_power_cap_kw	Hard cap; thermal risk is logged in the allocation reason.
M5	SOC/temperature sequence, candidate power	future_temperature_risk	Proactive trajectory cap before allocation.
M6	Safe action candidates, grid, M1-M5	strategy_value / Q value	Q ranks feasible strategies; it cannot revive a masked action.

5. Allocation and explainability

Capped water-filling repeatedly grants each open session a share of remaining station capacity proportional to P_i , never above its feasible cap. When a session reaches target SOC, its cap becomes zero and its power returns to the remaining open sessions on the next event. The UI records allocated kW, safe cap, priority, M1 need, readiness risk, M2 pressure, M4 thermal risk, and M5 future risk.

6. Invariants and deployment controls

Invariant	Implementation control
No unsafe allocation	Safety mask is evaluated before priority and water-filling.
No idle capacity when safe demand exists	Water-filling continues until site capacity or all caps are saturated.
No completed-session leakage	Target-SOC sessions receive a zero cap and trigger reallocation.
Traceability	Store model-version, input timestamp, signal vector, cap source, priority terms, and commanded kW.
Safe uncertainty handling	Treat stale or low-confidence telemetry as a lower cap or manual-review condition in production.

7. Production refinement checklist

Replace prototype estimators with versioned trained artifacts; calibrate thresholds per battery/charger OEM; validate BMS telemetry freshness; add command acknowledgement and fallback; log the complete allocation record; test against adversarial arrivals, lost telemetry, thermal rises, and feeder curtailment. This specification describes software behaviour, not a certified battery safety standard.

8. Scope note

PRISM-ANT is a technical design. It is not a promise of fire prevention, guaranteed battery life, or medical-vehicle response. Physical BMS, charger, electrical protection, OEM limits, and emergency procedures remain required.